

Commensurate Lerch Calculus

Finite cyclotomic closure, unequal-center expansions,
harmonic remainders, and Stieltjes–Gamma identities

Research continuation for Vladimir Reshetnikov’s ProveIt programme
Mathematical derivations and computational checks with OpenAI assistance

10 October 2026

Abstract

We extend reciprocal Lerch convolution from equal logarithmic scales to arbitrary commensurate positive scales, at every depth and every independent complex order. After a necessary scale normalization and alignment of the affine resolvents, the convolution depends only on the total order. A finite pole-lattice calculation reduces it to Lerch functions, and at the central point to pole-free Hurwitz-zeta combinations. We prove all mixed logarithmic moment identities, derive all-index Stieltjes and log-Gamma consequences, and give a normally convergent expansion for unequal affine centers, including every independent order derivative.

The resulting two-scale family includes $\int_0^\infty \log(1+y) \log(1+y^{-2}) \, dy/y = \pi G - 3\zeta(3)/8$, balanced unequal harmonic-remainder products, and explicit Gamma antiderivatives. A separate residue parity theorem works even for incommensurate scales; for example every central three-factor logistic convolution is $\pi^2 \sum_j q_j^{-2}/12$. Two precise rigidity statements identify when pure order addition and a common-step elementary kernel representation are possible. These statements do not assert arithmetic independence or unrestricted special-function nonreducibility.

The article supplies full analytic proofs, executable exact checks, and numerical diagnostics. It resolves the commensurate-dilation and unequal-shift-expansion research questions in the preceding reciprocal report, in the explicitly stated domains. It does not resolve the canonical Gaussian S_6 or revised S_8 conjectures, and makes no blanket claim of historical priority for individual identities.

Source anchor. Targeted canonical reading at repository commit

338a3a3877c337f981072061c16bc8ee1d87155d, together with the complete accessible source of the preceding reciprocal report. The incoming ZIP listing was inspected; its binary members were not audited. No repository files were changed.

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1 Research target and the new exact families

The ProveIt manuscript already contains a substantial theory of polylogarithmic identities, generalized harmonic sums, Hurwitz order jets, Stieltjes distributions, and Gamma correlations [1]. The preceding *Reciprocal Lerch Convolutions* report establishes entire-order addition, finite central-factorial formulas, and all logarithmic moments for equal logarithmic scales [2]. Its proposed research questions explicitly ask for commensurate dilations and for a normally convergent unequal-shift expansion with independent order jets. These are our principal targets. Re-proving an equal-scale identity is not counted as a new result.

The central distinction is between a spectral order and a logarithmic scale. If the argument y is replaced by y^q , the bilateral transform changes its cosecant period and also introduces a factor q^s . That factor cannot be ignored. Once it is removed, the new periods can be combined into a finite pole lattice exactly when their ratios are rational. A separate expansion handles unequal affine centers without pretending that a finite integer-order partial fraction identity can be differentiated in its integer index.

1.1 Two accessible consequences

For arbitrary complex s, t , write $W = s + t$. One result proved below is

$$\boxed{\int_0^\infty \text{Li}_s(-y) \text{Li}_t(-y^{-2}) \frac{dy}{y} = 2^{t-1} \pi \beta(W) - 2^{-s-1} W \eta(W+1).} \quad (1.1)$$

Here β is the Dirichlet beta function and η the Dirichlet eta function. Both sides are entire in the two orders, and the integral is ordinary and absolutely convergent. In particular, with $G = \beta(2)$ denoting Catalan's constant,

$$\int_0^\infty \log(1+y) \log(1+y^{-2}) \frac{dy}{y} = \pi G - \frac{3}{8} \zeta(3). \quad (1.2)$$

The two orders in (1.1) do not enter symmetrically. The normalization restoring pure order addition is multiplication of the integral by 2^{-t} .

A different consequence does not require rationally related scales. For any positive real q_1, q_2, q_3 ,

$$\boxed{\int_{\mathbb{R}^2} \frac{1}{1+e^{-q_1 x}} \frac{1}{1+e^{-q_2 y}} \frac{1}{1+e^{q_3(x+y)}} dx dy = \frac{\pi^2}{12} (q_1^{-2} + q_2^{-2} + q_3^{-2}).} \quad (1.3)$$

This is one case of an all-depth negative-integer parity theorem. It illustrates that the absence of a common pole lattice does not rule out exceptional exact central evaluations.

1.2 Conventions

We use the standard Hurwitz and Lerch functions [3]:

$$\zeta(v, a) = \sum_{n=0}^{\infty} (n+a)^{-v}, \quad \Phi(z, v, a) = \sum_{n=0}^{\infty} \frac{z^n}{(n+a)^v},$$

initially in their convergence domains and then by analytic continuation. Throughout, $\text{Re } a > 0$ unless another domain is explicitly stated; the powers use the logarithm on the right half-plane. The rising factorial is $(v)_j = v(v+1) \cdots (v+j-1)$, with $(v)_0 = 1$. A superscript $[h]$ means an order derivative. A superscript (h) on ψ means an ordinary argument derivative of the digamma function.

Our generalized Stieltjes convention is

$$\zeta(1+v, a) = \frac{1}{v} + \sum_{n=0}^{\infty} \frac{(-1)^n}{n!} \gamma_n(a) v^n, \quad \gamma_0(a) = -\psi(a). \quad (1.4)$$

We repeatedly use the entire functions

$$\mathcal{E}_a(v) = v\zeta(v+1, a), \quad \mathcal{E}_a(0) = 1, \quad (1.5)$$

$$\eta(v, a) = \Phi(-1, v, a) = 2^{-v} \left[\zeta\left(v, \frac{a}{2}\right) - \zeta\left(v, \frac{a+1}{2}\right) \right]. \quad (1.6)$$

In particular $\eta(v) = \eta(v, 1)$, while $\beta(v) = 4^{-v}[\zeta(v, 1/4) - \zeta(v, 3/4)]$. The singularities in these formulas are filled in before differentiation.

1.3 Relationship to classical methods and remaining conjectures

Euler's beta integral, Gamma reflection, convolution, pole summation, and Bell-polynomial differentiation are classical tools; the use of real convolution for Fermi–Dirac-type integrals is explicitly represented in Selvaggi and Selvaggi [5]. We do not claim to have invented these methods. The contribution here is the all-complex-order commensurate-scale closure, its precise hypotheses, pole-free normalization, and the derived families of identities and independent-center continuations.

The inspected canonical README records S_4 as proved and S_6 and the revised S_8 candidate as conjectural. Those statuses are unchanged here. A small residual is not an equality proof. The two research questions we answer are the analytic continuation problems just described, not those specific arithmetic reduction conjectures. Complete literature priority for every special value has not been established.

2 Aligned profiles and entire-order addition

For $y > 0$ define

$$F_{a,s}(y) = y\Phi(-y, s, a), \quad \ell(x) = \frac{e^x}{1+e^x}. \quad (2.1)$$

Thus $F_{1,s}(y) = -\text{Li}_s(-y)$ and $F_{a,0}(e^x) = \ell(x)$. For a positive real scale q and $\text{Re } A > -q$, set

$$h_{q,A,s}(x) = q^{-s} F_{1+A/q,s}(e^{qx}), \quad g_q(p) = \frac{\pi}{q} \csc \frac{\pi p}{q}. \quad (2.2)$$

We call A the affine center. It is $q(a-1)$ in the original Lerch shift, not a itself.

Lemma 2.1 (Weighted entire dependence). *Let $q > 0$, $\text{Re } A > -q$, and*

$$\max(0, -\text{Re } A) < c < q. \quad (2.3)$$

Then $s \mapsto e^{-cx} h_{q,A,s}(x)$ is entire with values in $L^1(\mathbb{R}) \cap L^\infty(\mathbb{R})$. The assertion holds after any fixed multiplication by a power of x , any fixed order or center derivative, and locally uniformly in A . The profiles are smooth in x and obey

$$(\partial_x + A)h_{q,A,s} = h_{q,A,s-1}. \quad (2.4)$$

Proof. For $\text{Re } s > 0$, the Lerch integral and the change of variable from its Laplace parameter to qu give

$$h_{q,A,s}(x) = \frac{1}{\Gamma(s)} \int_0^\infty u^{s-1} e^{-Au} \ell(q(x-u)) du. \quad (2.5)$$

Every derivative of ℓ is ℓ times a polynomial in ℓ . Consequently $|\ell^{(j)}(v)| \leq C_j \ell(v)$ on the real line. The function $e^{-cx} \ell(qx)$ is bounded and integrable precisely for $0 < c < q$. Multiplying (2.5) by e^{-cx} and writing $v = x - u$ separates the norm estimate into this logistic norm and

$$\int_0^\infty u^{\operatorname{Re} s - 1} e^{-(c + \operatorname{Re} A)u} du.$$

Both are finite. On compact subsets of the initial parameter domain, derivatives insert only powers of u , $\log u$, and smooth Gamma factors; these are controlled by slightly larger integrable majorants.

Integration by parts in u , initially for $\operatorname{Re} s > 1$, proves (2.4). For an arbitrary compact set of orders, choose an integer N with $\operatorname{Re}(s + N) > 0$ and define $h_{q,A,s} = (\partial_x + A)^N h_{q,A,s+N}$. The integration-by-parts identity makes these definitions consistent on overlaps. They agree with the convergent Lerch series for $x < 0$ and hence with the specified continuation. The derivative estimates for ℓ prove the norm-holomorphic extension. Finally, choose neighboring admissible weights $c \pm \varepsilon$. The bound $|x|^k \leq C_{k,\varepsilon} e^{\varepsilon|x|}$ proves every polynomially weighted assertion. Cauchy's formula in the parameters supplies all fixed parameter derivatives. $\square$

Theorem 2.2 (Scaled bilateral transform). *Under (2.3), for every $s \in \mathbb{C}$,*

$$\int_{\mathbb{R}} e^{-px} h_{q,A,s}(x) dx = g_q(p)(p + A)^{-s}, \quad \operatorname{Re} p = c. \quad (2.6)$$

The power uses the right-half-plane logarithm of $p + A$.

Proof. For $\operatorname{Re} s > 0$, Fubini is justified by the preceding proof. The transform of (2.5) factors into $(p + A)^{-s}$ and

$$\int_{\mathbb{R}} e^{-px} \ell(qx) dx = \frac{1}{q} \int_0^\infty \frac{y^{-p/q}}{1+y} dy = \frac{1}{q} \Gamma(p/q) \Gamma(1 - p/q) = g_q(p).$$

The last step is Gamma reflection [4]. Both sides are entire in s , so the identity extends to all orders. In particular there is no remaining $\Gamma(s)$ factor and no remaining q^s factor. $\square$

Let $\mathbf{q} = (q_1, \dots, q_r)$, $q_j > 0$, and $q_* = \min_j q_j$. For $r \geq 2$ put $x_r = \mu - \sum_{j < r} x_j$ and define

$$\mathcal{C}_{\mathbf{q},A}(W; \mu) = \int_{\mathbb{R}^{r-1}} \prod_{j=1}^r h_{q_j,A,s_j}(x_j) dx_1 \cdots dx_{r-1}, \quad W = \sum_j s_j. \quad (2.7)$$

For $r = 1$ the notation means evaluation of the single profile at μ . That the integral depends only on W is part of the following theorem.

Theorem 2.3 (Entire-order addition at arbitrary positive scales). *If $\operatorname{Re} A > -q_*$, (2.7) is absolutely convergent for all independent complex orders, depends only on their sum, and is entire in that sum. With $G_{\mathbf{q}}(p) = \prod_j g_{q_j}(p)$, it equals*

$$\mathcal{C}_{\mathbf{q},A}(W; \mu) = \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} e^{p\mu} G_{\mathbf{q}}(p)(p + A)^{-W} dp, \quad \max(0, -\operatorname{Re} A) < c < q_*. \quad (2.8)$$

Every fixed mixed order derivative can be moved inside the ordinary convolution integral. The contour expression is holomorphic for $|\operatorname{Im} \mu| < \pi \sum_j q_j^{-1}$.

Proof. Set $u_j(x) = e^{-cx} h_{q_j,A,s_j}(x)$. On the constraint hyperplane the weight removed from the product is the constant $e^{c\mu}$. Thus the absolute integral is at most $e^{c\mu} \|u_r\|_\infty \prod_{j < r} \|u_j\|_1$. Lemma 2.1 gives convergence and locally uniform bounds for all derivatives. The product of Fourier transforms is $G_{\mathbf{q}}(p)(p + A)^{-W}$, by Theorem 2.2. Along the vertical line this decays like $\exp(-\pi |\operatorname{Im} p| \sum_j q_j^{-1})$ times a fixed power of $1 + |p|$ on compact parameter sets. Fourier inversion therefore applies. The convolution is continuous, so equality holds pointwise. The same decay proves the claimed complex μ strip and all holomorphic dependence. $\square$

Write $K_q(\mu) = \mathcal{C}_{q,A}(0; \mu)$; it is independent of A and is the convolution of the scaled logistic functions. An equivalent representation, initially for $\operatorname{Re} W > 0$, is

$$\mathcal{C}_{q,A}(W; \mu) = \frac{1}{\Gamma(W)} \int_0^\infty u^{W-1} e^{-Au} K_q(\mu - u) du. \quad (2.9)$$

It follows either by the Dirichlet simplex integral in (2.5) or by its bilateral transform. The entire continuation in W , not the improper integral on the right, is used when $\operatorname{Re} W \leq 0$.

Scaling is also exact. For $\lambda > 0$,

$$\mathcal{C}_{\lambda q, \lambda A}(W; \mu) = \lambda^{1-r-W} \mathcal{C}_{q,A}(W; \lambda \mu). \quad (2.10)$$

Indeed $h_{\lambda q, \lambda A, s}(x) = \lambda^{-s} h_{q, A, s}(\lambda x)$, and the $r - 1$ integration variables supply the remaining power.

3 A finite pole lattice and pole-free Lerch closure

Assume now that the positive scales are commensurate: there exists $Q > 0$ with $d_j = Q/q_j \in \mathbb{N}$. For rational scales one may choose

$$Q = \frac{\operatorname{lcm}(\operatorname{num} q_1, \dots, \operatorname{num} q_r)}{\operatorname{gcd}(\operatorname{den} q_1, \dots, \operatorname{den} q_r)}. \quad (3.1)$$

This is a common multiple of the scales, not a common denominator. Put

$$\sigma = (-1)^{\sum_j d_j}, \quad \mathcal{P} = \bigcup_{j=1}^r \{q_j, 2q_j, \dots, Q\}, \quad d_\rho = \#\{j : \rho/q_j \in \mathbb{Z}\}. \quad (3.2)$$

Then $G_q(p + Q) = \sigma G_q(p)$, and every positive pole is a translate of a member of $\mathcal{P}$. Define its finite Laurent data by

$$G_q(\rho + z) = \sum_{k=1}^{d_\rho} a_{\rho,k} z^{-k} + O(1). \quad (3.3)$$

Lemma 3.1 (Elementary commensurate kernel). *For $\mu < 0$,*

$$K_q(\mu) = - \frac{\sum_{\rho \in \mathcal{P}} \sum_{k=1}^{d_\rho} a_{\rho,k} e^{\rho \mu} \frac{\mu^{k-1}}{(k-1)!}}{1 - \sigma e^{Q\mu}}. \quad (3.4)$$

If $\sigma = 1$, the simple Laurent coefficients satisfy

$$\sum_{\rho \in \mathcal{P}} a_{\rho,1} = 0. \quad (3.5)$$

Formula (3.4) continues to all real μ , with its removable value at zero understood when $\sigma = 1$.

Proof. Take the inverse transform of G_q on $0 < c < q_*$. For $\mu < 0$ close the contour to the right. Choose the closing vertical lines at a fixed nonpole residue class modulo Q . The trigonometric denominators are then uniformly separated from zero near the real axis, and give exponential decay at large imaginary part. The closing vertical integral tends to zero by $e^{\mu \operatorname{Re} p}$; the horizontal integrals vanish as their heights tend to infinity. The closure is clockwise, so the sign of the residue sum is negative.

At $p = \rho + nQ$, the coefficient of $(p - \rho - nQ)^{-k}$ is $\sigma^n a_{\rho,k}$. The residue of its product with $e^{p\mu}$ is $\sigma^n a_{\rho,k} e^{(\rho+nQ)\mu} \mu^{k-1} / (k-1)!$. The convergent geometric sum over $n \geq 0$ proves (3.4).

When $\sigma = 1$, integrate G_q around a rectangle whose two vertical edges differ by Q , enclosing one positive period of poles. The vertical integrals cancel by periodicity, and the horizontal ones vanish by exponential decay. The residue theorem gives (3.5). In (3.4) the numerator therefore vanishes at zero, so its possible real singularity is removable. The original inverse transform is holomorphic in a strip around the real axis. Analytic continuation from $\mu < 0$ along the real axis proves the last assertion. Possible more distant complex zeros of the denominator are not being declared removable here. $\square$

Theorem 3.2 (Finite Lerch closure at every complex order). *Let $\operatorname{Re} A > -q_*$ and $b_\rho = (A + \rho)/Q$. For $\mu < 0$ and $W \in \mathbb{C}$,*

$$\mathcal{C}_{q,A}(W; \mu) = - \sum_{\rho \in \mathcal{P}} \sum_{k=1}^{d_\rho} a_{\rho,k} e^{\rho\mu} \sum_{j=0}^{k-1} \frac{(-1)^j \mu^{k-1-j} (W)_j}{j! (k-1-j)! Q^{W+j}} \cdot \Phi(\sigma e^{Q\mu}, W + j, b_\rho). \quad (3.6)$$

The whole expression continues to the real axis. If $\sigma = 1$ and $\mu > 0$, its two Lerch boundary values agree. At $\mu = 0$ the following central expression is interpreted in its pole-free form below:

$$\mathcal{C}_{q,A}(W; 0) = \sum_{\rho \in \mathcal{P}} \sum_{k=1}^{d_\rho} \frac{(-1)^k a_{\rho,k}}{(k-1)! Q^{W+k-1}} (W)_{k-1} \Phi(\sigma, W + k - 1, b_\rho). \quad (3.7)$$

Proof. For $\operatorname{Re} W > 0$, insert (3.4) into (2.9). Expand $(\mu - u)^{k-1}/(k-1)!$ in powers of u , and use

$$\frac{1}{1 - \sigma e^{Q(\mu-u)}} = \sum_{n=0}^{\infty} \sigma^n e^{nQ\mu - nQu}.$$

The absolute convergence is immediate for $\mu < 0$ from $\operatorname{Re}(A + \rho) > 0$. Each Laplace integral is $\Gamma(W + j)/(A + \rho + nQ)^{W+j}$. Dividing by $\Gamma(W)$ produces $(W)_j$, proving (3.6) in the initial half-plane. Since the Lerch argument has modulus less than one, the right side is entire in W . Theorem 2.3 proves equality everywhere.

Initially with $\operatorname{Re} W$ sufficiently large, taking $\mu \rightarrow 0^-$ is justified in the summed integrals and leaves only $j = k - 1$. This gives (3.7). Its extension to all orders follows from the pole-free identities in Corollary 3.3.

For the real continuation when $\sigma = 1$, continue the equality from negative μ through the upper and lower parts of the holomorphic strip. The Lerch jump at $z = e^{Q\mu} > 1$ is, up to the common boundary-orientation sign,

$$\frac{2\pi i}{\Gamma(W + j)} e^{-b_\rho Q\mu} (Q\mu)^{W+j-1}.$$

For $\operatorname{Re}(W + j) > 0$, this jump follows by indenting the Lerch Euler integral at its simple pole $u = Q\mu$; its residue is precisely the displayed coefficient. The general order follows by continuation. After multiplication by the factors in (3.6), the ρ and j dependence in the exponential and in Q cancels. For fixed $k \geq 2$, the remaining coefficient is proportional to

$$\sum_{j=0}^{k-1} \frac{(-1)^j}{j! (k-1-j)!} = 0.$$

For $k = 1$, the total remaining coefficient is proportional to $\sum_\rho a_{\rho,1} = 0$. This initially proves zero jump away from the Gamma exceptional points and then everywhere by order continuation. Thus both boundary values coincide with the analytic ordinary convolution. For $\sigma = -1$ there is no positive-real Lerch cut to cross. $\square$

Corollary 3.3 (Entire central normalization). *If $\sigma = -1$, every summand of (3.7) is entire, with $\Phi(-1, v, b) = \eta(v, b)$. If $\sigma = 1$, choose any fixed reference b_* with positive real part. Replace the combined $k = 1$ terms by*

$$-Q^{-W} \sum_{\rho \in \mathcal{P}} a_{\rho,1} [\zeta(W, b_\rho) - \zeta(W, b_*)]. \quad (3.8)$$

For every $k \geq 2$ make the replacement

$$(W)_{k-1} \zeta(W + k - 1, b_\rho) = (W)_{k-2} \mathcal{E}_{b_\rho}(W + k - 2). \quad (3.9)$$

All these new summands are entire in W and their sum is the desired central convolution.

Proof. The residues of the two Hurwitz functions in a difference cancel. At $W = 1$ its value is $\psi(b_*) - \psi(b_\rho)$ by (1.4). Equation (3.5) makes subtraction of the reference function legitimate. For $k \geq 2$, the last factor of $(W)_{k-1}$ is $W + k - 2$, giving (3.9). The function $\mathcal{E}$ is entire by its removable normalization. These observations prove the claim, including every nonpositive-integer resonance. $\square$

The number of terms is finite independently of the orders. It depends only on the scales and pole multiplicities. No search for an integer relation among numerical special values enters the proof.

4 Exact Gamma and Bell formulas for the Laurent coefficients

The finite closure becomes an explicit algorithm once the coefficients $a_{\rho,k}$ are known. For a fixed pole put

$$I_\rho = \{j : \rho/q_j \in \mathbb{Z}\}, \quad J_\rho = \{1, \dots, r\} \setminus I_\rho, \quad d = |I_\rho|.$$

Define the complete exponential Bell polynomials by

$$\exp \left(\sum_{n \geq 1} u_n \frac{z^n}{n!} \right) = \sum_{n \geq 0} \mathcal{B}_n(u_1, \dots, u_n) \frac{z^n}{n!}. \quad (4.1)$$

Thus $\mathcal{B}_0 = 1$, $\mathcal{B}_1 = u_1$, $\mathcal{B}_2 = u_1^2 + u_2$, and $\mathcal{B}_3 = u_1^3 + 3u_1u_2 + u_3$.

Theorem 4.1 (Finite coefficient construction). *The leading coefficient at ρ is*

$$a_{\rho,d} = (-1)^{\sum_{j \in I_\rho} \rho/q_j} \prod_{j \in J_\rho} \frac{\pi}{q_j} \csc \frac{\pi \rho}{q_j}. \quad (4.2)$$

For $n \geq 1$ let

$$\begin{aligned} u_n &= \mathbf{1}_{2|n} 2(n-1)! \zeta(n) \sum_{j \in I_\rho} q_j^{-n} \\ &\quad + \sum_{j \in J_\rho} q_j^{-n} \left\{ \psi^{(n-1)}(\rho/q_j) + (-1)^n \psi^{(n-1)}(1 - \rho/q_j) \right\}, \end{aligned} \quad (4.3)$$

where the first line is zero for odd n , so it never calls for $\zeta(1)$. Then, for $1 \leq k \leq d$,

$$a_{\rho,k} = \frac{a_{\rho,d}}{(d-k)!} \mathcal{B}_{d-k}(u_1, \dots, u_{d-k}). \quad (4.4)$$

Only $u_1, \dots, u_{d-1}$ are needed. For rational scales, $a_{\rho,k}/\pi^{r-k}$ is a real cyclotomic algebraic number.

Proof. Each polar factor has leading term $(-1)^{\rho/q_j}/z$, proving (4.2). The normalized local germ $z^d G_{\mathbf{q}}(\rho + z)/a_{\rho,d}$ is analytic and equals one at zero; its local logarithm is consequently unambiguous. For a polar factor its logarithm is

$$\log \frac{\pi z/q_j}{\sin(\pi z/q_j)} = \sum_{h=1}^{\infty} \frac{\zeta(2h)}{h} \left(\frac{z}{q_j} \right)^{2h}.$$

This follows from the sine product, or from Gamma reflection and the local Gamma series. Its n th derivative at zero is the first line of (4.3). For a nonpolar factor use

$$g_{q_j}(\rho + z) = q_j^{-1} \Gamma\left(\frac{\rho + z}{q_j}\right) \Gamma\left(1 - \frac{\rho + z}{q_j}\right).$$

Differentiating its local logarithm gives the second line. The arguments $1 - \rho/q_j$ here may be negative, but are not integers; the meromorphic polygamma values are well defined. No global branch of their separate Gamma logarithms is required. Exponentiating the normalized local logarithm gives (4.4).

For the final assertion one can avoid transcendental cancellation entirely. Expand the sine factors directly at rational multiples of π , invert their finite Taylor polynomials, and multiply to order $d - 1$. The trigonometric constants are real cyclotomic algebraic numbers, and the Taylor powers supply exactly π^{r-k} . $\square$

The two constructions in this proof are independent computational routes. The supplied code uses the logarithmic Bell recurrence in `pole_table`, and direct reciprocal-sine Taylor multiplication in `direct_local_table`. Exact comparisons are made after symbolic simplification, not after floating-point evaluation of polygamma terms.

4.1 A genuinely mixed three-scale example

For $\mathbf{q} = (1, 2, 3)$, a common period is $Q = 6$ and $\sigma = -1$. The complete table is as follows; blank entries mean zero or absent higher pole order.

ρ	$a_{\rho,1}$	$a_{\rho,2}$	$a_{\rho,3}$
1	$-\sqrt{3}\pi^2/9$		
2	$-2\pi^2/27$	$-2\sqrt{3}\pi/9$	
3	0	$-\pi/2$	
4	$2\pi^2/27$	$-2\sqrt{3}\pi/9$	
5	$\sqrt{3}\pi^2/9$		
6	$-49\pi^2/216$	0	-1

Together with (3.7), this is an explicit finite identity for every total order and every $\text{Re } A > -1$, using alternating Hurwitz functions at $(A + \rho)/6$. No additional infinite sum is hidden in the coefficient construction. At $W = 0$ every $k \geq 2$ term vanishes because $(0)_{k-1} = 0$, and $\eta(0, b) = 1/2$. Thus

$$K_{(1,2,3)}(0) = -\frac{1}{2} \sum_{\rho} a_{\rho,1} = \frac{49\pi^2}{432}. \quad (4.5)$$

The independent parity theorem in Section 12 gives exactly the same value without constructing this table.

The coefficients also obey a useful scaling check. If all scales and poles are multiplied by $\lambda > 0$, then

$$a_{\lambda\rho,k}^{\lambda\mathbf{q}} = \lambda^{k-r} a_{\rho,k}^{\mathbf{q}}. \quad (4.6)$$

This follows at once from $G_{\lambda\mathbf{q}}(p) = \lambda^{-r} G_{\mathbf{q}}(p/\lambda)$ and is included in the exact verification suite.

5 The complete integer two-scale identity

The general table construction has a particularly short closed form for scales 1 and m , where m is any positive integer. Write

$$\sigma_m = (-1)^{m+1}, \quad c_r^{(m)} = (-1)^{r+1} \csc \frac{\pi r}{m} \quad (1 \leq r < m). \quad (5.1)$$

When $m = 1$ all sums over r below are empty. For $\operatorname{Re} A > -1$, define the *unnormalized* integral

$$I_m(s, t; A) = \int_0^\infty F_{1+A, s}(y) F_{1+A/m, t}(y^{-m}) \frac{dy}{y}. \quad (5.2)$$

The normalized quantity is $D_m(W; A) = \mathcal{C}_{(1, m), A}(W; 0) = m^{-t} I_m(s, t; A)$.

Theorem 5.1 (All integer scale ratios and all complex orders). *For $s, t \in \mathbb{C}$, $W = s + t$, and $\operatorname{Re} A > -1$,*

$$\begin{aligned} I_m(s, t; A) = m^{-s-1} & \left[\pi \sum_{r=1}^{m-1} c_r^{(m)} \Phi \left(\sigma_m, W, \frac{A+r}{m} \right) \right. \\ & \left. + \sigma_m W \Phi \left(\sigma_m, W+1, 1 + \frac{A}{m} \right) \right]. \end{aligned} \quad (5.3)$$

The integral is ordinary, absolutely convergent, and entire in s, t . For odd m , the last term is $\mathcal{E}_{1+A/m}(W)$, and the simple Hurwitz sum is grouped into differences to remove its pole at $W = 1$.

Proof. Use $Q = m$. The poles at $r = 1, \dots, m-1$ are simple, with

$$a_{r,1} = (-1)^r \frac{\pi}{m} \csc \frac{\pi r}{m}.$$

At $\rho = m$ both factors are polar, and $a_{m,2} = (-1)^{m+1} = \sigma_m$, $a_{m,1} = 0$. Substitution in (3.7) gives the normalized formula with prefactor m^{-W-1} . Multiplication by m^t gives (5.3). Entire dependence and absolute convergence were proved before the residue calculation. For odd m , $c_{m-r}^{(m)} = -c_r^{(m)}$, so $\sum_r c_r^{(m)} = 0$ and the simple-pole cancellation is explicit. $\square$

5.1 Scales two and three

At $A = 0$ the functions in the integrand are ordinary polylogarithms. For $m = 2$, the two reductions $\eta(W, 1/2) = 2^W \beta(W)$ and $\eta(W, 1) = \eta(W)$ give (1.1). For $m = 3$, let χ_{-3} be the real nonprincipal character modulo three and set

$$L(W, \chi_{-3}) = 3^{-W} [\zeta(W, 1/3) - \zeta(W, 2/3)].$$

Then

$$\boxed{\int_0^\infty \operatorname{Li}_s(-y) \operatorname{Li}_t(-y^{-3}) \frac{dy}{y} = \frac{2\pi 3^{t-1}}{\sqrt{3}} L(W, \chi_{-3}) + 3^{-s-1} \mathcal{E}_1(W).} \quad (5.4)$$

For $s = t = 1$ this yields

$$\int_0^\infty \log(1+y) \log(1+y^{-3}) \frac{dy}{y} = \frac{2\pi}{\sqrt{3}} L(2, \chi_{-3}) + \frac{2}{9} \zeta(3). \quad (5.5)$$

The first two numerical values are

$$\begin{aligned} I_2(1, 1; 0) &= 2.426819442923313294752678355954754053578 \dots, \\ I_3(1, 1; 0) &= 3.101375395830302048787723234392638 \dots \end{aligned}$$

They were checked against direct real-line logarithmic quadrature; they are diagnostics, not the proof.

5.2 Zero-order resonance and an explicit normalization obstruction

At zero orders one obtains

$$I_2(0, 0; 0) = \frac{\pi}{4}, \quad I_3(0, 0; 0) = \frac{1}{3} + \frac{2\pi}{9\sqrt{3}}. \quad (5.6)$$

The $1/3$ in the second expression is the value $3^{-1}\mathcal{E}_1(0)$. Substituting $W = 0$ in a separated factor $W\zeta(W+1)$ and declaring it zero would lose this term.

Even at the smallest nonzero total order the raw unequal-scale integral fails pure order addition:

$$I_2(1, 0; 0) = \frac{5\pi^2}{48}, \quad I_2(0, 1; 0) = \frac{5\pi^2}{24}. \quad (5.7)$$

For instance, use $\beta(1) = \pi/4$ and $\eta(2) = \pi^2/12$ in (1.1). These two values have the same total order but differ by a factor of two. The normalized integrals agree, as Theorem 2.3 requires. This is an exact counterexample to a tempting extension of the old equal-scale statement, not an error attributed to that statement itself.

For $m = 1$, Theorem 5.1 reduces to the preceding report's identity $I_1(s, t; A) = \mathcal{E}_{1+A}(s+t)$. This limiting consistency is useful, but the equal-scale case is prior work, not a new claim.

6 Balanced harmonic remainders and their order derivatives

The scale alignment also identifies the correct unequal truncation lengths. For $N \geq 0$ put

$$\mathcal{R}_{N,s}(y) = \text{Li}_s(-y) - \sum_{k=1}^N \frac{(-y)^k}{k^s}. \quad (6.1)$$

The empty sum is zero. For $a > 0$ and $\sigma \in \{1, -1\}$ write

$$H_{N,\sigma}^{(v)}(a) = \sum_{k=0}^{N-1} \frac{\sigma^k}{(k+a)^v}. \quad (6.2)$$

This finite function is entire in v . At $a = 1$, $\sigma = 1$, it is the usual generalized harmonic number $H_N^{(v)}$.

Theorem 6.1 (Unequal-scale harmonic-remainder product). *For all integers $m \geq 1$, $N \geq 0$, and all complex s, t , with $W = s + t$,*

$$\begin{aligned} \int_0^\infty \mathcal{R}_{mN,s}(y) \mathcal{R}_{N,t}(y^{-m}) \frac{dy}{y} &= I_m(s, t; 0) \\ &- m^{-s-1} \left[\pi \sum_{r=1}^{m-1} c_r^{(m)} H_{N,\sigma_m}^{(W)} \left(\frac{r}{m} \right) + \sigma_m W H_{N,\sigma_m}^{(W+1)}(1) \right]. \end{aligned} \quad (6.3)$$

The integral is absolutely convergent and entire in its two orders. Every independent order derivative can be passed through it.

Proof. The elementary shift of the Lerch series, first for $0 < y < 1$ and then by continuation along $y > 0$, gives

$$F_{N+1,s}(y) = -(-y)^{-N} \mathcal{R}_{N,s}(y). \quad (6.4)$$

Take $A = mN$ in (5.2). The two factors in (6.4) have powers y^{-mN} and y^{mN} , so they cancel; their signs give

$$\int_0^\infty \mathcal{R}_{mN,s}(y) \mathcal{R}_{N,t}(y^{-m}) \frac{dy}{y} = \sigma_m^N I_m(s, t; mN).$$

This also proves convergence and all order-holomorphic assertions by Theorem 5.1 and the weighted lemma.

The finite shift formula is

$$\Phi(\sigma, v, a + N) = \sigma^N [\Phi(\sigma, v, a) - H_{N,\sigma}^{(v)}(a)] \quad (\sigma = \pm 1).$$

It follows directly by removing the first N terms; at $\sigma = 1$ it is continued through the usual Hurwitz pole in the combined formula. All arguments in (5.3) increase by N when $A = mN$. The resulting factor σ_m^N cancels the preceding sign, proving (6.3). All harmonic corrections are finite entire functions of the orders, so no additional continuation issue remains. $\square$

An explicit example is

$$\begin{aligned} \int_0^\infty \left[-\log(1+y) + y - \frac{y^2}{2} \right] \left[-\log(1+y^{-2}) + y^{-2} \right] \frac{dy}{y} \\ = \pi G - \frac{3}{8}\zeta(3) - \pi + \frac{1}{2}. \end{aligned} \quad (6.5)$$

This is the case $m = 2$, $N = 1$, $s = t = 1$. The two brackets must be kept together. Expanding the product and integrating its monomials separately would introduce divergent integrals and would not prove the identity. The unequal truncation lengths 2 and 1 are dictated by the scales; they are not interchangeable with two equal truncation lengths.

6.1 Logarithmic harmonic derivatives

For every $h \geq 0$,

$$\partial_v^h H_{N,\sigma}^{(v)}(a) = (-1)^h \sum_{k=0}^{N-1} \frac{\sigma^k \log^h(k+a)}{(k+a)^v}. \quad (6.6)$$

Consequently the derivatives of (6.3) give explicit identities for products of polylogarithmic order derivatives and finite harmonic-logarithmic corrections, with no new infinite coefficients. The prefactor m^{-s-1} contributes powers of $-\log m$ and must be differentiated as well.

For example, the first s derivative of the correction bracket in (6.3), evaluated at any pair (s, t) , is

$$\begin{aligned} m^{-s-1} \left\{ \pi \sum_r c_r^{(m)} [\partial_W H_{N,\sigma_m}^{(W)}(r/m) - \log m H_{N,\sigma_m}^{(W)}(r/m)] \right. \\ \left. + \sigma_m [(1 - W \log m) H_{N,\sigma_m}^{(W+1)}(1) + W \partial_W H_{N,\sigma_m}^{(W+1)}(1)] \right\}. \end{aligned}$$

The t derivative has no derivative of the prefactor. Thus the explicit harmonic identities retain the same scale asymmetry as the raw polylogarithm integral. Arbitrarily high derivatives follow from the finite Leibniz rule rather than an unproved extrapolation.

7 Finite closure of every mixed logarithmic moment

The total-order theorem concerns the unweighted convolution. Polynomial weights in the logarithmic variables still have finite closure, even though they distinguish the independent orders. For a multi-index $\ell = (\ell_1, \dots, \ell_r)$ of nonnegative integers, put $L = \sum_j \ell_j$ and define

$$\mathcal{M}_\ell(\mathbf{s}; A; \mu) = \int_{\mathbb{R}^{r-1}} \prod_{j=1}^r x_j^{\ell_j} h_{q_j, A, s_j}(x_j) d\mathbf{x}, \quad x_r = \mu - \sum_{j < r} x_j. \quad (7.1)$$

The measure denotes the first $r - 1$ variables as before.

Theorem 7.1 (A terminating mixed-moment algorithm). *Let the scales be commensurate and $\operatorname{Re} A > -q_*$. Then (7.1) is absolutely convergent and entire in all orders. For $0 \leq D \leq L$, form the meromorphic trigonometric function*

$$G_D(p; \mathbf{s}) = \sum_{\substack{0 \leq d_j \leq \ell_j \\ \sum_j d_j = D}} \prod_{j=1}^r \binom{\ell_j}{d_j} (s_j)_{d_j} (-\partial_p)^{\ell_j - d_j} g_{q_j}(p). \quad (7.2)$$

Compute the finite Laurent table of each G_D on the same period Q . Apply formula (3.6), or its pole-free central version, to that table with order $W + D$. Summing the answers over D gives exactly $\mathcal{M}_\ell$. The pole set remains $\mathcal{P}$, and no pole order exceeds $r + L - D$. Every fixed mixed order derivative of this finite formula is valid.

Proof. Multiplication by x^{ℓ_j} corresponds to $(-\partial_p)^{\ell_j}$ on the bilateral transform. Lemma 2.1 justifies these operations and the integrals. The Leibniz rule gives

$$(-\partial_p)^{\ell_j} [g_{q_j}(p)(p + A)^{-s_j}] = \sum_{d=0}^{\ell_j} \binom{\ell_j}{d} (s_j)_d (-\partial_p)^{\ell_j - d} g_{q_j}(p)(p + A)^{-s_j - d}.$$

Multiplying and grouping by D yields $\sum_D G_D(p; \mathbf{s})(p + A)^{-W - D}$. Every derivative of a cosecant has the same signed period as the cosecant, so $G_D(p + Q; \mathbf{s}) = \sigma G_D(p; \mathbf{s})$. Differentiation creates no new poles and increases pole orders by at most the number of derivatives. The total is at most $r + L - D$. All these functions retain exponential vertical decay.

The residue proof of Lemma 3.1 and Theorem 3.2 uses precisely these properties, not a special factorization into undifferentiated cosecants. Thus it applies to each G_D . In the periodic case the simple residues still sum to zero by the period rectangle. The same pole-removal and branch-jump proofs therefore remain valid. The resulting expression is a finite sum of entire functions of the orders with polynomial coefficient factors. The weighted holomorphy proves that it also equals every differentiated ordinary integral claimed in the statement. $\square$

7.1 A convenient differential-polynomial implementation

For a single scale set $\kappa_q = (\pi/q)^2$ and define

$$\begin{aligned} R_0(s; T, U; \kappa) &= 1, \\ R_{\ell+1} &= (T + sU)R_\ell + (\kappa + T^2)\partial_T R_\ell + U^2\partial_U R_\ell. \end{aligned} \quad (7.3)$$

With $T = (\pi/q) \cot(\pi p/q)$ and $U = (p + A)^{-1}$ one has

$$(-\partial_p)^\ell [g_q(p)(p + A)^{-s}] = g_q(p)(p + A)^{-s} R_\ell(s; T, U; \kappa_q). \quad (7.4)$$

Indeed $g'_q = -Tg_q$, $T' = -(\kappa_q + T^2)$ and $U' = -U^2$; the product rule proves the recurrence. The first cases are

$$\begin{aligned} R_1 &= T + sU, \\ R_2 &= 2T^2 + 2sTU + (s)_2 U^2 + \kappa, \\ R_3 &= 6T^3 + 6sT^2U + 3(s)_2 TU^2 + (s)_3 U^3 + 5\kappa T + 3s\kappa U. \end{aligned}$$

Different scales have different T and κ values. Setting every κ to π^2 would silently revert to equal scales. Formula (7.2) avoids that error and provides an unambiguous finite normal form before pole extraction.

There are also exact consistency constraints. At every μ ,

$$\sum_{j=1}^r \mathcal{M}_{e_j}(\mathbf{s}; A; \mu) = \mu \mathcal{C}_{q,A}(W; \mu). \quad (7.5)$$

This follows directly from $\sum_j x_j = \mu$ in the ordinary integral. Unlike the equal-scale first-moment formulas, the individual terms need not be a fixed $1/r$ share. The identity is a check on their sum, not a claim that scale asymmetry disappears.

8 All-index Stieltjes identities and Gamma specializations

8.1 Ordinary products of normalized order jets

Define the normalized order jets

$$\mathcal{L}_{q,n}^A(x) = \partial_s^n h_{q,A,s}(x)|_{s=0}. \quad (8.1)$$

The normalization q^{-s} is part of this definition; its derivatives contribute powers of $-\log q$. For a multi-index $\mathbf{n} = (n_1, \dots, n_r)$, $M = \sum_j n_j$, Theorem 2.3 gives the ordinary product law

$$\int_{\mathbb{R}^{r-1}} \prod_{j=1}^r \mathcal{L}_{q_j, n_j}^A(x_j) d\mathbf{x} = \mathcal{C}_{\mathbf{q}, A}^{[M]}(0; 0), \quad x_r = -\sum_{j < r} x_j. \quad (8.2)$$

It is absolutely convergent. In particular, even at unequal scales the answer depends on the distribution of order derivatives only through their total M . It is not a Hermitian product, so no positivity of these Stieltjes expressions is implied.

Here are the exact rules that turn every central closure into Stieltjes and zeta derivatives:

$$\mathcal{E}_a^{[h]}(0) = \begin{cases} 1, & h = 0, \\ (-1)^{h-1} h \gamma_{h-1}(a), & h \geq 1, \end{cases} \quad (8.3)$$

$$\mathcal{E}_a^{[h]}(v) = v \zeta^{[h]}(v+1, a) + h \zeta^{[h-1]}(v+1, a) \quad (v \neq 0). \quad (8.4)$$

The second term in (8.4) is absent when $h = 0$. The first identity is just the Laurent convention (1.4) multiplied by v .

For example, when $\sigma = 1$ an explicit all-index coefficient formula for the right side of (8.2) is

$$\begin{aligned} M! [z^M] Q^{-z} & \left\{ - \sum_{\rho} a_{\rho,1} [\zeta(z, b_{\rho}) - \zeta(z, b_*)] \right. \\ & \left. + \sum_{\rho} \sum_{k=2}^{d_{\rho}} \frac{(-1)^k a_{\rho,k}}{(k-1)! Q^{k-1}} (z)_{k-2} \mathcal{E}_{b_{\rho}}(z+k-2) \right\}. \end{aligned} \quad (8.5)$$

This is finite coefficient extraction from explicitly given analytic germs. The only Stieltjes constants in it arise from (8.3); the other terms involve ordinary zeta order derivatives at zero or positive integers. For $\sigma = -1$, use instead

$$M! [z^M] Q^{-z} \sum_{\rho} \sum_{k=1}^{d_{\rho}} \frac{(-1)^k a_{\rho,k}}{(k-1)! Q^{k-1}} (z)_{k-1} \eta(z+k-1, b_{\rho}). \quad (8.6)$$

These formulas, together with finite product differentiation, are explicit at every index; they do not require a recurrence fitted to low-order data.

8.2 Generalized harmonic coefficients

For $j \geq 1$ the exact local identity

$$(z)_j = (j-1)!z \exp \left(\sum_{h=1}^{\infty} \frac{(-1)^{h-1}}{h} H_{j-1}^{(h)} z^h \right) \quad (8.7)$$

follows by taking the logarithm of $\prod_{k=1}^{j-1} (1+z/k)$. Consequently, for $1 \leq l \leq j$,

$$\partial_z^l (z)_j \Big|_{z=0} = l(j-1)! \mathcal{B}_{l-1}(H_{j-1}, -H_{j-1}^{(2)}, 2!H_{j-1}^{(3)}, \dots). \quad (8.8)$$

Derivatives of order greater than j vanish because $(z)_j$ is a polynomial. Equations (8.5)–(8.8) therefore express every jet by finite generalized-harmonic polynomials, zeta derivatives, and Stieltjes constants. At other integer resonances, the finite polynomial should be shifted and expanded directly; dividing by a vanishing Pochhammer factor is unnecessary and unsafe.

8.3 A short all-index formula for two scales

Put $b_r = (A+r)/m$ and $b_m = 1 + A/m$. Define $e_0(b) = 1$ and $e_h(b) = (-1)^{h-1} h \gamma_{h-1}(b)$ for $h \geq 1$. For odd m , Theorem 5.1 gives

$$D_m^{[M]}(0; A) = \frac{1}{m} \sum_{h=0}^M \binom{M}{h} (-\log m)^{M-h} \left[\pi \sum_{r=1}^{m-1} c_r^{(m)} \zeta^{[h]}(0, b_r) + e_h(b_m) \right]. \quad (8.9)$$

For even m , the corresponding formula is

$$D_m^{[M]}(0; A) = \frac{1}{m} \sum_{h=0}^M \binom{M}{h} (-\log m)^{M-h} \left[\pi \sum_{r=1}^{m-1} c_r^{(m)} \eta^{[h]}(0, b_r) - h \eta^{[h-1]}(1, b_m) \right], \quad (8.10)$$

where the last term is zero when $h = 0$. These identities follow from the ordinary product rule applied to $D_m(W; A) = m^{-W-1}[\dots]$ in (5.3); all differentiated integrals are justified by the weighted lemma.

The apparently separate eta jets at one are themselves finite Stieltjes differences:

$$\eta^{[h]}(1, b) = \frac{1}{2} \sum_{j=0}^h \binom{h}{j} (-\log 2)^{h-j} (-1)^j \left[\gamma_j \left(\frac{b}{2} \right) - \gamma_j \left(\frac{b+1}{2} \right) \right]. \quad (8.11)$$

Indeed the two Hurwitz poles in (1.6) cancel before multiplication by 2^{-v} ; expansion at $v = 1$ then gives the formula. For $h = 0$ it agrees with

$$\eta(1, b) = \frac{1}{2} \left[\psi \left(\frac{b+1}{2} \right) - \psi \left(\frac{b}{2} \right) \right]. \quad (8.12)$$

This check fixes the sign of every Stieltjes difference.

8.4 Quarter-Gamma and third-Gamma product identities

The classical identity $\zeta^{[1]}(0, b) = \log \Gamma(b) - \frac{1}{2} \log(2\pi)$ [3] gives

$$\eta^{[1]}(0, b) = \log \Gamma(b/2) - \log \Gamma((b+1)/2) - \frac{1}{2} \log 2. \quad (8.13)$$

Here and below the logarithm of Gamma is the holomorphic one on the right half-plane, real on the positive axis. At $A = 0$, two explicit consequences are

$$D_2^{[1]}(0; 0) = \frac{\pi}{2} \log \frac{\Gamma(1/4)}{\Gamma(3/4)} - \frac{\pi + 1}{2} \log 2, \quad (8.14)$$

$$D_3^{[1]}(0; 0) = \frac{2\pi}{3\sqrt{3}} \log \frac{\Gamma(1/3)}{\Gamma(2/3)} - \frac{2\pi \log 3}{9\sqrt{3}} + \frac{\gamma - \log 3}{3}. \quad (8.15)$$

By (8.2), the first value is, for example,

$$\int_{\mathbb{R}} \mathcal{L}_{1,1}^0(x) \mathcal{L}_{2,0}^0(-x) dx = \int_{\mathbb{R}} \mathcal{L}_{1,0}^0(x) \mathcal{L}_{2,1}^0(-x) dx.$$

The derivative inside the second integral includes the scale normalization, so this does not contradict (5.7).

One order higher, (5.4) in normalized form gives

$$D_3^{[2]}(0; 0) = \frac{2\pi}{3\sqrt{3}} L^{[2]}(0, \chi_{-3}) + \frac{(\log 3)^2 - 2\gamma \log 3 - 2\gamma_1}{3}. \quad (8.16)$$

Thus an ordinary product of first order jets, with two different scales, contains a Dirichlet L -derivative and the first Stieltjes constant in one exact formula. General A in (8.9) and (8.10) replaces these by generalized Stieltjes constants and shifted Gamma values.

9 Argument derivatives and all anchored primitive orders

Order differentiation is distinct from differentiating the center or the logarithmic argument. The contour formula makes the latter two operations equally explicit:

$$\partial_A \mathcal{C}_{q,A}(W; \mu) = -W \mathcal{C}_{q,A}(W + 1; \mu), \quad (9.1)$$

$$(\partial_\mu + A) \mathcal{C}_{q,A}(W; \mu) = \mathcal{C}_{q,A}(W - 1; \mu). \quad (9.2)$$

Both follow by differentiating (2.8); its exponential vertical majorant justifies each derivative. They also follow from the corresponding profile identities. Iteration gives

$$\partial_A^d \mathcal{C}_{q,A}(W; \mu) = (-1)^d (W)_d \mathcal{C}_{q,A}(W + d; \mu). \quad (9.3)$$

Every mixed spectral derivative can be applied to these identities. The Pochhammer factor must then be differentiated too.

9.1 First and higher anchored antiderivatives

For an anchor A_0 in $\operatorname{Re} A > -q_*$,

$$\int_{A_0}^A \mathcal{C}_{q,u}(W; \mu) du = \frac{\mathcal{C}_{q,A_0}(W - 1; \mu) - \mathcal{C}_{q,A}(W - 1; \mu)}{W - 1}. \quad (9.4)$$

The integral is along any path in that half-plane. The right side is filled in at $W = 1$ by

$$\mathcal{C}_{q,A_0}^{[1]}(0; \mu) - \mathcal{C}_{q,A}^{[1]}(0; \mu), \quad (9.5)$$

since $\mathcal{C}_{q,A}(0; \mu) = K_q(\mu)$ is independent of A . At $\mu = 0$, this already gives a finite Gamma and polygamma primitive of the weight-one convolution.

The same mechanism supplies every primitive order. Define

$$\begin{aligned} T_d(W; A, A_0; \mu) &= \mathcal{C}_{q,A}(W - d; \mu) \\ &\quad - \sum_{j=0}^{d-1} \frac{(A - A_0)^j}{j!} (-1)^j (W - d)_j \mathcal{C}_{q,A_0}(W - d + j; \mu). \end{aligned} \quad (9.6)$$

Theorem 9.1 (All anchored primitives, including resonance). *For $d \geq 1$,*

$$\boxed{\int_{A_0}^A \frac{(A-u)^{d-1}}{(d-1)!} \mathcal{C}_{q,u}(W; \mu) du = \frac{(-1)^d T_d(W; A, A_0; \mu)}{(W-d)_d}}. \quad (9.7)$$

The right side is entire in W after removing its apparent poles at $W = 1, \dots, d$. At $W = k$ in that range its value is

$$\frac{(-1)^k}{(k-1)!(d-k)!} \partial_W T_d(W; A, A_0; \mu)|_{W=k}. \quad (9.8)$$

All order derivatives of these primitive identities are valid.

Proof. Apply the ordinary Taylor formula with integral remainder, in the variable A , to $\mathcal{C}_{q,A}(W-d; \mu)$ at A_0 . Its j th derivative is given by (9.3). Its d th derivative is $(-1)^d (W-d)_d \mathcal{C}_{q,A}(W; \mu)$. This proves (9.7) when the polynomial denominator is nonzero. The left side is entire in W , by local uniform holomorphy on a compact integration path. Therefore all apparent poles are removable. The polynomial $(W-d)_d = \prod_{j=1}^d (W-j)$ has simple zeros, with its derivative at $W = k$ equal to $(-1)^{d-k} (k-1)!(d-k)!$. Taking the quotient limit gives (9.8). Local uniform holomorphy and Cauchy's formula give every order derivative. $\square$

This formulation fixes all integration constants by the anchor and its vanishing derivatives through order $d-1$. It is stronger than writing an unanchored formal primitive and leaving its constants ambiguous at resonant orders.

9.2 Explicit Gamma primitives of the two-scale family

For $D_m(W; A) = \mathcal{C}_{(1,m),A}(W; 0)$, define $b_r = (A+r)/m$. When m is odd, put

$$P_m(A) = \frac{1}{m} \left[\psi(b_m) - \pi \sum_{r=1}^{m-1} c_r^{(m)} \log \Gamma(b_r) \right]. \quad (9.9)$$

When m is even, put

$$P_m(A) = \frac{1}{m} \left[\eta(1, b_m) - \pi \sum_{r=1}^{m-1} c_r^{(m)} \log \frac{\Gamma(b_r/2)}{\Gamma((b_r+1)/2)} \right]. \quad (9.10)$$

Then in both cases

$$P'_m(A) = D_m(1; A), \quad \int_{A_0}^A D_m(1; u) du = P_m(A) - P_m(A_0). \quad (9.11)$$

To check the odd case directly, use $\psi'(b) = \zeta(2, b)$ and $\sum_r c_r^{(m)} \zeta(1, b_r) = -\sum_r c_r^{(m)} \psi(b_r)$, where the sum is understood after its residue cancellation. For the even case use $\partial_b \eta(1, b) = -\eta(2, b)$ and

$$\partial_b \log \frac{\Gamma(b/2)}{\Gamma((b+1)/2)} = -\eta(1, b).$$

Both derivatives reproduce the $W = 1$ case of the normalized pair formula, including its m^{-2} factor. This is also an independent check of the sign in (9.5).

Subsequent center derivatives of (9.9) and (9.10) are finite polygamma expressions. The all-index Stieltjes formulas in Section 8, followed by (9.7), give their anchored order-jet primitives. Thus differentiation and integration are both covered without replacing a pole-cancelled expression by its singular separate pieces.

10 Unequal centers: normal convergence through every order jet

The preceding finite identities share one affine center A . For independent centers A_j , define

$$\mathcal{U}_{q,A}(\mathbf{s}; \mu) = \int_{\mathbb{R}^{r-1}} \prod_{j=1}^r h_{q_j, A_j, s_j}(x_j) d\mathbf{x}. \quad (10.1)$$

We require a common weight:

$$\max_j(0, -\min \operatorname{Re} A_j) < c < q_*. \quad (10.2)$$

This condition is stronger than the separate requirements $\operatorname{Re} A_j > -q_j$ when the scales are unequal; it must not be dropped. It holds, in particular, if all $\operatorname{Re} A_j > -q_*$. The weighted proof of Theorem 2.3 applies to give

$$\mathcal{U}_{q,A}(\mathbf{s}; \mu) = \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} e^{p\mu} G_q(p) \prod_{j=1}^r (p + A_j)^{-s_j} dp. \quad (10.3)$$

It is entire in its independent orders; no integer-order restriction is being analytically differentiated.

Choose a real reference A with $A + c > 0$ and set $\delta_j = A_j - A$. Suppose

$$\rho_0 = \max_j \frac{|\delta_j|}{A + c} < 1. \quad (10.4)$$

Define explicit polynomial coefficients by

$$\prod_{j=1}^r (1 + \delta_j z)^{-s_j} = \sum_{N=0}^{\infty} h_N(\mathbf{s}; \boldsymbol{\delta}) z^N, \quad (10.5)$$

$$h_N(\mathbf{s}; \boldsymbol{\delta}) = \sum_{n_1 + \dots + n_r = N} \prod_{j=1}^r \frac{(-\delta_j)^{n_j} (s_j)_{n_j}}{n_j!}. \quad (10.6)$$

In particular $h_0 = 1$, $h_1 = -\sum_j s_j \delta_j$, and

$$h_2 = \frac{1}{2} \sum_j s_j (s_j + 1) \delta_j^2 + \sum_{i < j} s_i s_j \delta_i \delta_j.$$

Theorem 10.1 (Normally convergent unequal-center closure). *Under (10.2) and (10.4),*

$$\boxed{\mathcal{U}_{q,A}(\mathbf{s}; \mu) = \sum_{N=0}^{\infty} h_N(\mathbf{s}; \boldsymbol{\delta}) \mathcal{C}_{q,A}(W + N; \mu), \quad W = \sum_j s_j.} \quad (10.7)$$

The series is normally convergent on compact subsets where a strict version of (10.4) holds, jointly in the independent orders, the centers, and μ in a closed substrip of $|\operatorname{Im} \mu| < \pi \sum_j q_j^{-1}$. Every fixed mixed order or center derivative may be taken term by term. Center collisions are regular. If the scales are commensurate, every coefficient function $\mathcal{C}$ on the right has the finite pole-free Hurwitz or Lerch form already proved.

Proof. On the contour, $|p + A| \geq A + c$, so $|\delta_j/(p + A)| \leq \rho_0 < 1$ uniformly. The chosen logarithms satisfy

$$\log(p + A_j) = \log(p + A) + \log\left(1 + \frac{\delta_j}{p + A}\right).$$

This is first clear as δ_j moves from zero inside the strict disk, and continues there because both $p + A$ and $p + A_j$ stay in the right half-plane. The binomial series therefore gives

$$\prod_j (p + A_j)^{-s_j} = (p + A)^{-W} \sum_{N \geq 0} h_N(\mathbf{s}; \boldsymbol{\delta}) (p + A)^{-N}.$$

On a compact set of orders, $|(s)_n/n!|$ is at most a fixed polynomial in n ; this follows directly from the Gamma ratio away from a finite initial set, with continuity covering nonpositive integer orders. The absolute binomial series is consequently bounded by a convergent polynomial-times-geometric series, uniformly along the whole contour. The remaining factors in (10.3) have an integrable exponential vertical majorant, locally uniformly in all parameters. Tonelli's theorem for the majorant and ordinary dominated convergence justify termwise contour integration, giving (10.7).

The same estimates hold on a slightly larger compact neighborhood of any strict parameter set. Cauchy's integral formula then proves normal convergence of every fixed derivative, not merely of the undifferentiated series. In particular no singularity is created when two δ_j coincide. The final assertion follows by substituting Theorem 3.2 into each term. $\square$

Corollary 10.2 (Every common-strip configuration has such an expansion). *For any fixed centers satisfying (10.2), a reference A satisfying (10.4) exists. One may choose a real number larger than zero and all the finitely many numbers*

$$\frac{|A_j|^2 - c^2}{2(c + \operatorname{Re} A_j)}. \quad (10.8)$$

Proof. The denominators are positive. The strict inequality $|A_j - A|^2 < (A + c)^2$ is exactly $|A_j|^2 - c^2 < 2A(c + \operatorname{Re} A_j)$, so the stated choice works for every j . $\square$

This proves the unequal-shift research question in the preceding reciprocal report [2, Further research, question 2] for its right-half-plane shifts, and extends it to unequal scales whenever the common strip exists. For equal scales $q_j = 1$, the old shifts are $a_j = 1 + A_j$; their conditions $\operatorname{Re} a_j > 0$ imply that such a common $c < 1$ can be chosen. Thus the original equal-scale question is fully covered, including all independent spectral derivatives.

The result should not be described as a finite reduction of arbitrary unequal centers: the outer series is generally infinite. The finite closure is retained *inside each term*. Theorem 11.1 below explains why generic unequal centers do not recover pure order addition.

11 Two precise rigidity statements

11.1 When can the independent orders add purely?

Theorem 11.1 (Affine-center alignment is necessary and sufficient). *Fix positive scales and centers with a common strip. The convolution function $\mu \mapsto \mathcal{U}_{\mathbf{q}, \mathbf{A}}(\mathbf{s}; \mu)$ depends on the orders only through $W = \sum_j s_j$, for all orders in a nonempty open set, if and only if $A_1 = \dots = A_r$. For the unnormalized profiles $F_{1+A_j/q_j, s_j}(e^{q_j x})$, pure addition as a function of μ also requires $q_1 = \dots = q_r$.*

Proof. Sufficiency in the normalized case is Theorem 2.3. For necessity, weighted convolution is in L^1 as well as continuous, so one can take its bilateral transform. If the function depends only on W , its derivative in the direction $\partial_{s_j} - \partial_{s_k}$ vanishes. By (10.3) and transform uniqueness, this gives

$$G_{\mathbf{q}}(p) \prod_l (p + A_l)^{-s_l} [\log(p + A_k) - \log(p + A_j)] = 0$$

on the common vertical line. The prefactors never vanish there, so the logarithms agree. Exponentiation implies $p + A_k = p + A_j$, hence $A_k = A_j$. This holds for every pair.

In the unnormalized case there is an additional transform factor $\prod_j q_j^{s_j}$. The same derivative now gives $\log q_j - \log q_k - \log(p + A_j) + \log(p + A_k) = 0$. Thus $q_j(p + A_k) = q_k(p + A_j)$ on a line. Comparing the linear coefficients in p gives $q_j = q_k$, and then $A_j = A_k$. $\square$

The statement concerns the entire convolution *function* of μ , not a special numerical coincidence at one point or one order. It does not rule out exceptional identities at $\mu = 0$. Even with equal scales, unequal centers already fail in the simplest example:

$$\begin{aligned}\mathcal{U}_{(1,1),(0,1)}((1,0);0) &= \zeta(2), \\ \mathcal{U}_{(1,1),(0,1)}((0,1);0) &= \zeta(2) - 1.\end{aligned}$$

To verify these, the zero-order profile is independent of its center. The two integrals therefore reduce respectively to the equal-center weight-one formulas at $A = 0$ and $A = 1$.

11.2 What commensurability really characterizes

Consider the following explicitly restricted elementary class on $x < 0$: finite sums of functions

$$\frac{P_\nu(x)e^{\rho_\nu x}}{1 - \epsilon_\nu e^{Qx}}, \quad Q > 0, \quad |\epsilon_\nu| = 1, \quad (11.1)$$

where each P_ν is a polynomial, and the finite data ρ_ν , ϵ_ν and P_ν are constant in x . The same Q is used for every term. Complex exponents are allowed.

Theorem 11.2 (Common-step kernel rigidity). *For positive real scales $\mathbf{q}$, the logistic convolution $K_{\mathbf{q}}$ has a representation of the form (11.1) on $x < 0$ if and only if all scale ratios are rational.*

Proof. The sufficiency is (3.4), since commensurate scales have a positive common multiple Q . For necessity, suppose such a finite representation exists and fix $\delta > 0$. On $x < -\delta$, expand the denominators as geometric series. Integrating each term of its negative-half-line Laplace transform gives meromorphic functions of p whose poles lie in the finite union

$$\bigcup_{\nu} \{\rho_\nu + nQ : n \geq 0\}.$$

Polynomial factors only increase pole multiplicities. The upper cutoff $-\delta$ supplies exponential decay in n , so after omitting any finite initial terms the integrated series is normally convergent on compact subsets away from that pole set. This establishes the stated meromorphic pole restriction, regardless of any cancellations between terms of the finite representation.

The Laplace transform over $[-\delta, \infty)$ is holomorphic for $\operatorname{Re} p > 0$. Indeed the weighted convolution estimates with arbitrarily small positive weight show that $K_{\mathbf{q}}(x)$ grows more slowly than $e^{\varepsilon x}$ for every $\varepsilon > 0$ as $x \rightarrow +\infty$. The full transform agrees with $G_{\mathbf{q}}(p)$ on $0 < \operatorname{Re} p < q_*$, so meromorphic continuation forces every positive pole of $G_{\mathbf{q}}$ to belong to the finite union above.

But $G_{\mathbf{q}}$ has an uncanceled pole at every nq_j , $n \geq 1$: cosecant factors have no zeros, so another factor cannot remove it. For a fixed j , two of the infinitely many nq_j lie in the same one of the finitely many progressions. Subtracting them gives $(n - n')q_j = (k - k')Q$, hence $q_j/Q \in \mathbb{Q}$. This holds for each j , proving commensurability. $\square$

This theorem is a characterization of a fixed common-step elementary kernel class. It is *not* a claim that incommensurate integrals cannot be expressed by other special functions, cannot have isolated exact values, or yield arithmetically independent constants. The next section supplies abundant isolated exact values for arbitrary positive scales, emphasizing the distinction.

12 Universal parity identities beyond commensurate scales

At negative integer total orders the resolvent power becomes a polynomial. A local residue at zero then yields exact identities without a common period. Put

$$B_q(z) = z^r G_q(z) = \prod_{j=1}^r \frac{\pi z / q_j}{\sin(\pi z / q_j)}. \quad (12.1)$$

This is an even analytic function near zero with $B_q(0) = 1$.

Theorem 12.1 (Reflection-residue parity law). *Let $q_j > 0$ be arbitrary and let $N \geq 0$ be an integer. Then*

$$\begin{aligned} \mathcal{C}_{q,A}(-N; 0) - (-1)^{r+N} \mathcal{C}_{q,-A}(-N; 0) \\ = [z^{r-1}](z + A)^N B_q(z). \end{aligned} \quad (12.2)$$

The two ordinary integrals are covered simultaneously when $|\operatorname{Re} A| < q_*$. At these integer orders each side is a polynomial in A , so the identity also gives their polynomial continuation to all A . In particular, if $r + N$ is odd,

$$\boxed{\mathcal{C}_{q,0}(-N; 0) = \frac{1}{2}[z^{r-N-1}]B_q(z).} \quad (12.3)$$

A coefficient with negative index is zero.

Proof. At $W = -N$, the contour integrand is $G_q(p)(p + A)^N$. Choose $0 < c < q_*$. At integer order the expression is a polynomial in A , so the contour can be considered independently of any branch of a resolvent power. Move the vertical contour from c to $-c$. The only pole crossed is zero; horizontal edges vanish by exponential decay. If $I_c(A)$ denotes the integral on $\operatorname{Re} p = c$,

$$I_c(A) - I_{-c}(A) = \operatorname{Res}_{p=0} G_q(p)(p + A)^N.$$

Since $G_q(-p) = (-1)^r G_q(p)$, changing variable $p \mapsto -p$ in the second vertical integral gives $I_{-c}(A) = (-1)^{r+N} I_c(-A)$. The residue on the right is exactly $[z^{r-1}](z + A)^N B_q(z)$. On the common ordinary-integral domain, $I_c(\pm A) = \mathcal{C}_{q,\pm A}(-N; 0)$, proving (12.2); the polynomial continuation follows at once. At $A = 0$ and $r + N$ odd, the left side is twice the desired value, giving (12.3). $\square$

12.1 Odd-depth logistic integrals

For $r = 2h + 1$ and $N = 0$, the theorem gives

$$K_q(0) = \frac{1}{2}[z^{2h}] \prod_{j=1}^{2h+1} \frac{\pi z / q_j}{\sin(\pi z / q_j)}. \quad (12.4)$$

Every such value is a homogeneous polynomial in the inverse squared scales with rational coefficients times π^{2h} . To see both the rationality and the exact construction, use

$$\frac{u}{\sin u} = 1 + \frac{u^2}{6} + \frac{7u^4}{360} + \cdots,$$

whose coefficients are rational by formal division of the sine series. The first two nontrivial cases are

$$K_{(q_1, q_2, q_3)}(0) = \frac{\pi^2}{12} \sum_{j=1}^3 q_j^{-2}, \quad (12.5)$$

$$K_{(q_1, \dots, q_5)}(0) = \frac{\pi^4}{720} \left(7 \sum_{j=1}^5 q_j^{-4} + 10 \sum_{i < j} q_i^{-2} q_j^{-2} \right). \quad (12.6)$$

Equation (12.5) is the ordinary double integral (1.3). For $(q_1, q_2, q_3) = (1, \sqrt{2}, \sqrt{3})$, for instance, its value is $11\pi^2/72$, despite the absence of a common scale period. For $(1, 2, 3)$ it recovers $49\pi^2/432$ from the explicit Laurent table.

More generally,

$$\mathcal{C}_{(q_1, q_2), 0}(-1; 0) = \frac{1}{2}, \quad \mathcal{C}_{(q_1, q_2, q_3), 0}(-2; 0) = \frac{1}{2}. \quad (12.7)$$

If $r + N$ is odd and $N > r - 1$, the value is exactly zero. Such identities involve negative-order Lerch or polylogarithmic profiles, which can change sign, and so do not contradict positivity of the zero-order logistic convolution.

At nonzero A , equation (12.2) supplies a finite reflection identity instead of an evaluation of each side separately. For example, at $r = 2$, $N = 2$ the residue is $2A$, so

$$\mathcal{C}_{(q_1, q_2), A}(-2; 0) - \mathcal{C}_{(q_1, q_2), -A}(-2; 0) = 2A.$$

This exact reflection is independent of the two scales. The theorem makes no assertion that the complementary-parity central values have a comparable universal polynomial form.

13 Proof status, correction safeguards, and reproducible checks

13.1 What was actually inspected

The canonical manuscript README and its source-directory structure were retrieved through the connected GitHub repository. The branch commit recorded during this audit is

338a3a3877c337f981072061c16bc8ee1d87155d.

The repository's timestamp for that commit is 11 October 2026 at 01:59:45 UTC, which is 10 October in the user's Pacific timezone. The full accessible TeX source of *Reciprocal Lerch Convolutions* was retrieved from the user's Library and preserved locally for this research pass. Its profile definitions, transform and weighted-holomorphy proof, equal-scale closure, moment formulas, Stieltjes conventions, harmonic-remainder identity, and future questions are the relevant source material. That report records its own, earlier repository pin; it must not be confused with the pin above.

The current incoming directory listing includes the bilinear Mellin, cubic mixed-order, exact-resonance, independent-order, harmonic parity, harmonic shift, higher reflection, nonlinear transport, ordered Hurwitz, and unequal-twist packages. Their filenames and GitHub metadata were visible, but their binary ZIP members were not read in this pass. Accordingly this article does not assert that its results are absent from every incoming archive, and does not claim to have audited those archives. The source-audit artifact records this limitation explicitly. A filename alone is not mathematical evidence of either novelty or overlap.

13.2 Corrections and integration safeguards

No new error in the inspected equal-scale theorem is asserted here. The following are concrete corrections to plausible *extensions* and implementation mistakes, supported by exact formulas in this article.

Scale normalization. Replace a raw order-addition claim for unequal scales by the normalized profiles (2.2). The exact pair (5.7) refutes the unnormalized claim. Alignment means $q_j(a_j - 1) = A$, not equality of the original Lerch shifts a_j .

Pole cancellation before specialization. Store $\mathcal{E}_a(v)$ as an entire object with value one at zero, and store simple-pole Hurwitz sums as differences. The $1/3$ in (5.6) is lost by zero-times-pole substitution. In particular the expression $W\zeta(W+1)$ is not zero at $W=0$.

Periods and branches. The pole period is a common multiple of the positive scales. Using a common denominator instead generally misidentifies the pole lattice. At positive μ in the periodic case, use the entire combined Lerch expression; its zero jump was proved, not assumed. Independently choosing branch values and discarding imaginary parts is not the rule.

Independent orders and common strips. An integer-only partial fraction formula cannot be differentiated in its changing integer summation limit. The normal expansion in Theorem 10.1 supplies an actual holomorphic family. For unequal scales and centers, the common-strip hypothesis (10.2) is explicit; separate one-profile admissibility is not silently substituted for it.

Primitive constants and residue scope. Use anchored Taylor remainders for higher primitives. Their resonant quotients have specified removable values. The universal parity result is an equality of the stated ordinary integrals on their common domain, then a polynomial continuation at negative integer order; it is not a license to continue arbitrary complex-order integrals without tracking branches and convergence.

Arithmetic and numerical scope. A functional pole-lattice obstruction is not arithmetic independence. The current S_6 and revised S_8 statuses are unchanged. During exploratory numerical work, an ordinary Python expression 3^{-2} introduced a binary floating-point coefficient into an otherwise high-precision comparison; replacing it by exact or multiprecision arithmetic removed that discrepancy. The delivered numerical routines convert rational constants without such a binary-float intermediate. This was an implementation issue in the present exploratory work, not an error attributed to the repository.

13.3 Executed verification

The delivered completed runs comprise **307 exact symbolic assertions, 91 numerical diagnostics, and five negative controls**. The exact and numerical roles are deliberately separate.

Script/run	Exact	Numerical	Negative controls
<code>verify.py -part exact</code>	285	0	2
<code>verify.py -part numeric</code>	0	77	2
<code>verify_extra.py</code>	22	14	1
Total	307	91	5

The exact suite compares the Bell and direct reciprocal-sine algorithms for eighteen scale families, checks period residues and scale covariance, verifies the full printed (1, 2, 3) table, and checks the two-scale pole formulas, mixed-moment recurrence, harmonic Pochhammer coefficients, primitive resonance denominators, and low-depth universal parity polynomials. The corruption control changes an actual coefficient of the Laurent table and verifies that the independent construction rejects it. A separate domain control rejects a zero scale.

The numerical suite compares finite central closure with independent vertical-contour quadrature, including complex orders, complex centers, zero order, positive order, and negative-integer resonance. It also compares noncentral Lerch closure, ordinary logarithmic product integrals, weighted moments, Gamma primitives, Stieltjes jets, balanced harmonic remainders, unequal-center expansions, and irrational-scale parity values. Higher primitive tests compare

the closed formulas to 32-point Gauss–Legendre integration; their resonant order derivatives are evaluated by a separate 48-point Cauchy sum. The universal cubic formula is also checked against ordinary one-dimensional integrals after an elementary two-factor convolution.

The usual working precision is 55 decimal digits. The balanced-remainder integral uses 100 digits to control cancellation and a truncated real-line integration interval; its acceptance tolerance is 10^{-30} . The other default tolerances are 10^{-36} , with 10^{-42} for the unequal-center expansion tests. The finite quadrature and finite expansion errors are not interval-enclosed. An initial adaptive higher-primitive diagnostic exceeded the execution limit; the delivered fixed quadrature and Cauchy-sum checks completed successfully. All completed records, including actual residuals and software versions, are in `verification/*-results.json`.

These computations test the formulas and their implementation. They are not interval certificates, not proofs for all parameters, and not a proof-assistant formalization. The all-index and continuum assertions rest on the analytic proofs above.

13.4 Integration status

Claim	Status in this delivery
Commensurate-scale, all-depth, entire-order finite closure	Proved.
Every mixed logarithmic moment and independent order jet	Proved by a finite algorithm.
Unequal-center expansion in the common-strip domain	Proved, normally convergent.
All anchored primitive orders and their resonances	Proved.
Universal negative-integer parity for arbitrary positive scales	Proved.
Pure-order-addition and common-step kernel rigidity	Proved in the stated functional classes.
Canonical Gaussian S_6 and revised S_8	Not proved or disproved here.
Novelty relative to every unread incoming ZIP	Not established.
Historical priority for every special case	Not claimed.

All theorem and equation labels use the prefix `clc:` to reduce integration collisions. The package includes modular sources, a standalone source, a compiled PDF, reusable verification code, provenance notes, and a checksum manifest. No change to the canonical repository was made.

14 Further research questions and concrete next targets

The following questions concern further exact identities rather than inequalities or asymptotic error bounds. They also distinguish genuine remaining problems from the two continuation questions resolved here.

Q1. A compact multivariate tilt generating function. For the constraint $\sum_j x_j = \mu$, introduce independent factors $e^{\lambda_j x_j}$. Their transform becomes $\prod_j g_{q_j}(p - \lambda_j)(p + A - \lambda_j)^{-s_j}$. Find a compact representation in a specified multivariate Lerch or Lauricella family, with every collision and normalization explicit, that avoids an outer center-expansion series. The present moment theorem already determines every Taylor coefficient, but does not supply such a compact resummation. A characterization of its genuinely finite sectors would be more informative than an unqualified “closed form.”

- Q2. The complementary negative-order parity sector.** Theorem 12.1 evaluates the sector $r + N$ odd at $A = 0$. For $r + N$ even and incommensurate scales, seek exact representations in Barnes-type or multiple Hurwitz functions that retain the full scale symmetry. A correct formula should reduce to the finite commensurate closure when all scale ratios become rational, without assuming that the limiting pole multiplicities are unchanged.
- Q3. Minimal cyclotomic coordinates and conductor descent.** The finite tables live in explicit real cyclotomic algebras, but those algebras and the Hurwitz argument grids are often redundant. Determine an exact reduction algorithm for the coefficient field and the Hurwitz-coordinate conductor simultaneously. The target is a canonical functional reduction with exact certificates, not numerical independence of a chosen set of periods. The two-scale beta and character-three formulas are required base cases.
- Q4. Finite closure without pure order addition.** Alignment is characterized exactly for pure addition. It does not classify all finite identities with unequal centers. Positive integer orders permit finite partial fractions, while our arbitrary-order series is usually infinite. Classify exceptional affine-center and order subvarieties on which the outer series terminates or resumes finitely. Nonpositive integer orders, repeated centers, and rational center differences provide distinct test cases and should not be conflated.
- Q5. Beyond a single common-step elementary kernel class.** Theorem 11.2 allows one shared positive Q . Determine the exact obstruction when finitely many different denominator periods are allowed in the elementary representation. The uncanceled pole support alone may no longer suffice; the Laurent coefficient sequences along intersecting progressions carry additional information. Any stronger theorem should state its representational class precisely, and should remain compatible with the universal isolated parity values.
- Q6. Several independent multiplicative constraints.** Replace the single linear condition on logarithms by an integer matrix of constraints. The transform is then a product of cosecants and affine resolvents on a higher-dimensional space. Seek a finite exact polylogarithm/Hurwitz closure when the pole hyperplane arrangement is rational and the resolvents are aligned along the appropriate dual subspaces. A first nontrivial target is two independent constraints with four or five factors, including intersecting-pole residues and all independent order derivatives.
- Q7. Colored profiles and exact contour-crossing terms.** Replace the negative argument of the Lerch profile by a root-of-unity multiple, and allow complex logarithmic shifts. Derive the correct weighted strip and the finite residue or branch terms when a pole crosses the chosen contour. The jump cancellation of Theorem 3.2 is a special aligned case, not a general rule that all colored jumps vanish. Exact quarter-turn specializations could connect this extension to the manuscript's Gaussian identities.
- Q8. A canonical finite basis for mixed moments.** The algorithm in Theorem 7.1 is terminating but not claimed minimal. Determine all functional syzygies among its differentiated Laurent tables and produce a canonical reduced representative that respects center differentiation, scale covariance, and (7.5). The equal-scale central-factorial recurrence is an instructive subcase; unequal periods introduce new redundancies.
- Q9. Machine-checked analytic infrastructure.** Formalize weighted entire families, bilateral convolution inversion, and the finite period-residue argument in a proof assistant.

The exact coefficient tables and negative controls can be used as finite fixtures, but they do not replace a formal proof of contour deformation, normal convergence, or branch cancellation. The first useful formal milestone is the all-complex-order two-scale theorem with its removable resonances.

The commensurate-dilation problem and the unequal-shift order-jet expansion posed in [2] now have complete proofs in the stated settings. The next advances should go beyond those solved settings, while preserving their normalization and convergence checks. In particular, connecting these integral families to the unresolved canonical Gaussian reduction vectors would require a separate explicit identity or exact reduction certificate; it is not a consequence of finite closure alone.

References

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- [2] ProveIt research continuation, *Reciprocal Lerch Convolutions: Entire-order identities, central-factorial zeta reductions, all logarithmic moments, and Stieltjes–Gamma calculus*, 10 October 2026. User Library source: ProveIt_Reciprocal_Lerch_Convolutions_2026-10-10.tex. The report records the earlier repository anchor 0c740647c48baec7f0b4192723e977d69917dd52. In particular, see its weighted transform, equal-scale closure, and further research questions 2 and 5. The source hash is retained in the accompanying source-audit artifact.
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- [5] J. P. Selvaggi and J. A. Selvaggi, “The Application of Real Convolution for Analytically Evaluating Fermi-Dirac-Type and Bose-Einstein-Type Integrals,” *Journal of Complex Analysis* **2018** (2018), article 5941485. <https://doi.org/10.1155/2018/5941485>. Cited for the prior use of real convolution in this class of integrals, not as a source of the present all-scale theorems.